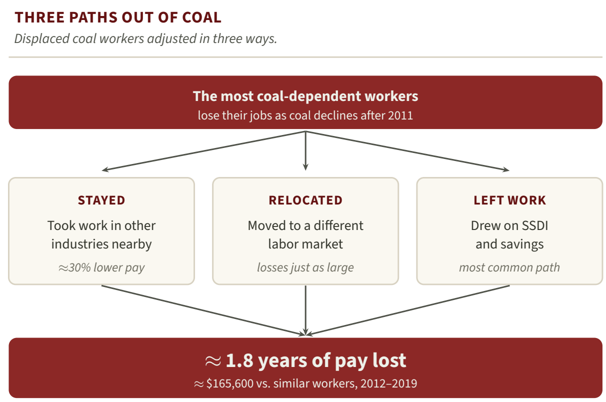
Coal's Decline Cost Workers Nearly Two Years of Pay

Lessons from "Transitional Costs and the Decline of Coal: Worker-Level Evidence" by Jonathan Colmer, Eleanor Krause, Eva Lyubich & John Voorheis, Working Paper, 2026.

When U.S. coal production and employment fell by more than half after 2011, the workers left behind never caught back up. The most coal-dependent workers lost close to two years' worth of earnings in the eight years that followed, and neither switching industries nor moving helped them recoup it.

BACKGROUND

Cheap natural gas from the fracking boom and falling wind and solar costs pushed coal out of the power sector, halving U.S. coal mining employment since its 2011 peak. Earlier research shows that coal *communities* lost jobs, earnings, and tax revenue, but community averages can hide what happened to individual workers. Some may have moved quickly into new jobs. Others may have borne heavy **transitional costs**: the lasting losses to earnings and employment that workers bear when an industry shrinks.



In a new working paper, Colmer et al. (2026) find that the workers most dependent on coal experienced significant earnings losses. The researchers assembled Census and IRS records covering nearly every person employed in U.S. coal mining between 2005 and 2021, following each worker's earnings, employment, and location year by year. That included any time they spent working outside coal, or not working at all.

THE CHALLENGE

Measuring what coal's decline cost workers is hard, because a career is shaped by many forces at once. Tracking coal workers' earnings after 2011 and calling a drop the true cost is misleading because aging, the broader economy, and ordinary job churn can all pull on those same paychecks. A raw decline can't be pinned on coal alone.

Comparing coal workers against everyone else also raises some concerns. Coal workers tend to be better paid, clustered in a few isolated regions, and different from other workers in schooling, so any earnings gap between the two groups could reflect those existing differences rather than what coal's decline did.

Colmer et al. compared the most coal-dependent workers (those in coal in all five years before the 2011 peak) with otherwise similar workers who had spent less time in coal, or none at all. The comparison rests on the assumption that the two groups' earnings would have moved together had coal not declined. That holds up here because coal's collapse came from national gas and renewables prices. It happened to these workers rather than because of them, so there is little reason to think the most coal-dependent group was already headed for greater losses.

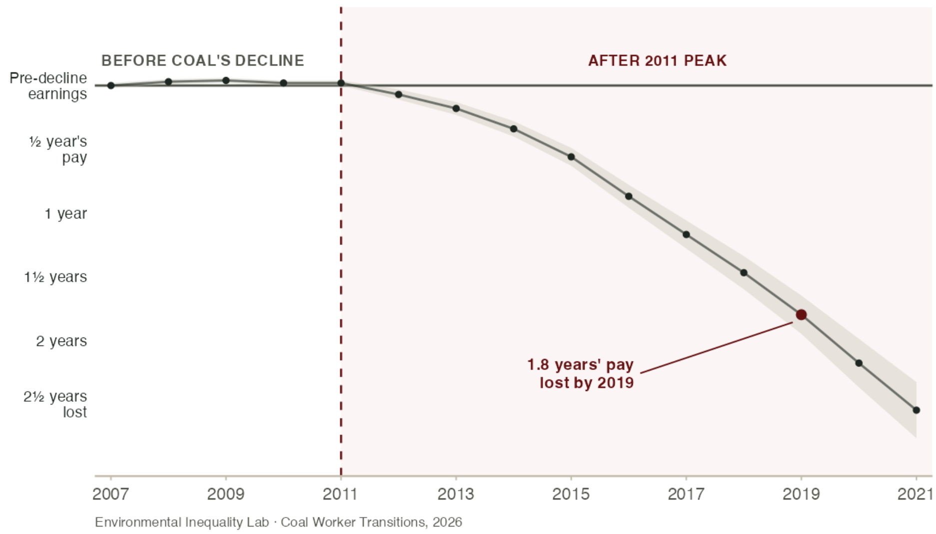
NEARLY TWO YEARS OF LOST EARNINGS, AND NO WAY BACK

Between 2012 and 2019, **the most coal-dependent workers lost cumulative earnings equal to roughly 1.8 times a full pre-decline year of pay**, about $165,600 for someone making $92,000 in 2011 (in 2019 dollars; see figure). The losses came through two channels at once: compared with similar workers, coal-dependent workers spent about five fewer months employed *and*, in the years they did work, earned roughly 19% less.

Four in five workers left coal. What they couldn't find was comparable pay. Coal paid about $13,000 a year more than other full-time work, and the industry was regionally isolated, so alternatives were scarce. Outside coal, they earned more than 30% less than comparable non-coal workers. Relocating didn't close the gap either. Most often, they didn't work at all. They leaned on Social Security Disability Insurance and drew down retirement savings, which replaced only a fraction of what was lost. **Coal's decline left a deep hole in workers' earnings that the labor market never filled.**

COAL WORKERS' EARNINGS FELL FURTHER BEHIND COMPARABLE WORKERS EVERY YEAR

Source: recreated from Colmer et al. (2026), Fig. 7a.



Note: Each point is the cumulative earnings gap between the most coal-dependent workers and otherwise similar workers who depended on coal less, measured in years of pre-decline pay. After coal's 2011 peak the two groups steadily diverge; by 2019 the gap reaches 1.8 years' pay. Had coal's decline cost these workers nothing, the line would have stayed flat at zero. Values digitized from the published figure.

IMPLICATIONS FOR POLICY

Coal is the first major U.S. fossil-fuel industry to go through a large-scale contraction, an early look at what falling demand could mean elsewhere in the carbon-intensive economy. The industries most exposed share coal's profile: high pay, few local alternatives, and a workforce whose skills are worth far less outside the industry. **Policy that assumes displaced workers will find comparable jobs is likely to be wrong** — these workers didn't, whether they stayed, switched industries, or moved.

The harder thing to replace was what they already knew how to do, not where they lived. Workers who left their local labor market fared no better than those who stayed, which points to skill mismatch rather than geography as the more binding constraint. How far that mismatch can be eased is an open question, though the researchers see room for cheaper responses than the disability benefits and retirement savings these workers fell back on. What coal makes clear is the price of leaving it in place: eight years on, the cost still sat with the workers.

FOR MORE INFORMATION

Contact Eleanor Krause (eleanor.krause@uky.edu) with questions about this research.

Colmer, Jonathan, Eleanor Krause, Eva Lyubich, and John Voorheis. 2026. "Transitional Costs and the Decline of Coal: Worker-Level Evidence." Working paper. Available at environmental-inequality-lab.org/energy.

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